Code (Line) Coverage Analysis Report

Lab 6 – Code and Branch Coverage | Ahana H Sharan | PES1UG24AM020

1. Objective & Method

Measure how many statements of food_delivery.py are executed by the tests, and add tests until line coverage is at least 90%.

2. Initial Coverage (starter test file)

The starter test_food_delivery.py has 2 tests (delivery_charge(2, 200, False) and apply_coupon(300, None, False)).

Stmts	Missed	Line Coverage	Missing lines
33	21	36%	6, 8, 11-14, 16, 23-29, 33-40

Untested: the out-of-range error, free delivery, medium/far distance and rain surcharges, every coupon path except None, and the whole estimated_time function.

3. Test Cases Added (CodeCoverageTest_food_delivery_application.py)

Test	Input	Expected	Lines covered
test_out_of_range	delivery_charge(0, 200, False)	ValueError	6
test_free_delivery_large_order_in_rain	delivery_charge(5, 500, True)	30	8, 16
test_medium_distance	delivery_charge(5, 200, False)	40	11-12
test_far_distance	delivery_charge(10, 200, False)	60	13-14
test_invalid_coupon	apply_coupon(300, 'FAKE', False)	ValueError	23-24
test_welcome_not_first_order	apply_coupon(300, 'WELCOME', False)	ValueError	25-26
test_coupon_order_below_minimum	apply_coupon(150, 'SAVE50', False)	150	27-28
test_coupon_applied	apply_coupon(300, 'SAVE50', False)	250	29
test_peak_hour_and_busy_restaurant	estimated_time(5, True, True)	65	33-40

4. Result

Stage	Tests	Stmts	Missed	Line Coverage
Before	2	33	21	36%
After	11 (all passed)	33	0	100%

```
============================================== 11 passed in 0.05s ==============================================
PS C:\Projects\SWE\L6_L7_20_food_delivery\Lab 6> python -m coverage report -m
Name                                Stmts   Miss  Cover   Missing
-----------------------------------------------------------------
CodeCoverageTest_food_delivery.py      27      0   100%
food_delivery.py                       33      0   100%
-----------------------------------------------------------------
TOTAL                                  60      0   100%
```

5. Analysis

• Line coverage went from **36% to 100%** (target ≥ 90%) by adding one test for each untested statement block.

• 100% line coverage does **not** mean every decision was tested. Running the same tests with --branch gives **95%** branch coverage, with 3 partial branches in estimated_time (34→36, 36→40, 38→40). Only the case where every condition is True was run, so the False sides were never taken.

• Those remaining paths are covered in the branch coverage report.